AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	18.2 / 3.06768 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-16 00:00 UTC
Location	29.6863, -95.2947
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 616	57.73 to 100; mean 89.1584	94.24 at 2026-06-15 23:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 616	22.7778 to 33.3333; mean 26.7453	26 at 2026-06-15 23:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 616	0 to 360; mean 120.211	20 at 2026-06-15 23:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 616	0 to 11.3178; mean 2.5889	2.05778 at 2026-06-15 23:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 143	5851.27 to 5926.57; mean 5888.9	5885.77 at 2026-06-16 00:00Z; at event time
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 143	32.0322 to 1802.53; mean 342.235	48.199 at 2026-06-16 00:00Z; at event time
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 143	52.1939 to 69.5435; mean 59.9716	60.9815 at 2026-06-16 00:00Z; at event time
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 143	1001.26 to 1014.43; mean 1008.97	1009.44 at 2026-06-16 00:00Z; at event time
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 143	16.785 to 19.498; mean 18.322	18.8384 at 2026-06-16 00:00Z; at event time
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 143	-4.71304 to 1.58078; mean -0.9417	-0.619447 at 2026-06-16 00:00Z; at event time
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 143	-1.42555 to 6.14875; mean 1.9996	-0.575546 at 2026-06-16 00:00Z; at event time
openaq	Ozone (ozone)	ppm	17 / 1110	-0.001 to 0.031; mean 0.0135	0.01 at 2026-06-16 00:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 212	5 to 110; mean 25.0377	14 at 2026-06-16 00:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	90 / 3111	-2.6 to 52; mean 6.9223	5.7 at 2026-06-16 00:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 355	3 to 100; mean 91.6113	100 at 2026-06-16 00:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 355	68 to 100; mean 90.5859	88 at 2026-06-16 00:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 355	0 to 24.39; mean 1.1861	0.1 at 2026-06-16 00:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 355	1005 to 1016; mean 1011.44	1011 at 2026-06-16 00:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 355	24.22 to 33.3; mean 26.8152	26.72 at 2026-06-16 00:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 355	0 to 358; mean 159.786	89 at 2026-06-16 00:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 355	0.07 to 5.67; mean 2.0365	2.07 at 2026-06-16 00:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	66 / 4758	0 to 46.2; mean 8.1691	8.7 at 2026-06-16 00:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 4 h 50 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 4 h 50 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	1 / 1	0.0259566 to 0.0259566; mean 0.026	0.0259566 at 2026-06-14 19:24Z; 28 h 35 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 4 h 50 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	1 / 1	0.000111617 to 0.000111617; mean 0.0001	0.000111617 at 2026-06-14 19:24Z; 28 h 35 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 4 h 50 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	1 / 1	2.03572e-05 to 2.03572e-05; mean 0	2.03572e-05 at 2026-06-14 19:24Z; 28 h 35 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 4 h 50 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 4 h 50 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	1 / 1	-6.10873e-06 to -6.10873e-06; mean -0	-6.10873e-06 at 2026-06-14 19:24Z; 28 h 35 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 4 h 50 min before event
tceq	Carbon monoxide (co)	ppm	3 / 213	0.1 to 1.1; mean 0.223	0.3 at 2026-06-16 00:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 790	-2.1 to 27; mean 5.6039	18.2 at 2026-06-16 00:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 219	-0.6 to 0.6; mean -0.1169	-0.5 at 2026-06-16 00:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-14 12Z 80.89, 18Z 69.98, 06-15 00Z 88.16, 06Z 92.9, 12Z 93.04, 18Z 90.65, 06-16 00Z 93.26, 06Z 97.13, 12Z 93.15, 18Z 86.17, 06-17 00Z 89.65, 06Z 95.39
asos	Air temperature (temperature)	06-14 12Z 28.76, 18Z 30.74, 06-15 00Z 27.65, 06Z 26.9, 12Z 25.98, 18Z 26.07, 06-16 00Z 25.96, 06Z 25.46, 12Z 25.49, 18Z 26.98, 06-17 00Z 26.19, 06Z 24.8
asos	Wind direction (wind_direction)	06-14 12Z 147, 18Z 176.3, 06-15 00Z 115, 06Z 124, 12Z 173.5, 18Z 106.5, 06-16 00Z 68.46, 06Z 67.17, 12Z 155.8, 18Z 106.6, 06-17 00Z 106, 06Z 101.3
asos	Wind speed (wind_speed)	06-14 12Z 3.611, 18Z 4.066, 06-15 00Z 2.229, 06Z 2.379, 12Z 2.744, 18Z 1.271, 06-16 00Z 1.533, 06Z 1.126, 12Z 3.255, 18Z 2.048, 06-17 00Z 2.815, 06Z 3.921
noaa_gfs	500-hPa geopotential height (gh_500)	06-14 12Z 5911, 18Z 5926, 06-15 00Z 5921, 06Z 5922, 12Z 5903, 18Z 5893, 06-16 00Z 5886, 06Z 5891, 12Z 5872, 18Z 5865, 06-17 00Z 5855, 06Z 5858, 12Z 5853
noaa_gfs	Planetary boundary layer height (pbl_height)	06-14 12Z 257.7, 18Z 1416, 06-15 00Z 468, 06Z 225.2, 12Z 264.2, 18Z 316.4, 06-16 00Z 147.4, 06Z 74.46, 12Z 157.9, 18Z 420.5, 06-17 00Z 286.1, 06Z 131.4, 12Z 283.8
noaa_gfs	Precipitable water (precipitable_water)	06-14 12Z 57.96, 18Z 59.03, 06-15 00Z 60.59, 06Z 59.99, 12Z 58.11, 18Z 64.91, 06-16 00Z 61.25, 06Z 62.82, 12Z 64.55, 18Z 63.2, 06-17 00Z 55.72, 06Z 56.3, 12Z 55.2
noaa_gfs	Surface pressure (surface_pressure)	06-14 12Z 1012, 18Z 1012, 06-15 00Z 1010, 06Z 1012, 12Z 1011, 18Z 1011, 06-16 00Z 1009, 06Z 1009, 12Z 1008, 18Z 1008, 06-17 00Z 1005, 06Z 1005, 12Z 1003
noaa_gfs	850-hPa temperature (t_850)	06-14 12Z 18.52, 18Z 18.04, 06-15 00Z 18.96, 06Z 19.25, 12Z 18.61, 18Z 17.21, 06-16 00Z 18.8, 06Z 18.41, 12Z 17.78, 18Z 17.53, 06-17 00Z 18.87, 06Z 18.04, 12Z 18.19
noaa_gfs	10-m eastward wind (u_10m)	06-14 12Z -1.28, 18Z -0.8351, 06-15 00Z -1.073, 06Z -0.4866, 12Z -0.3975, 18Z -1.02, 06-16 00Z -0.3913, 06Z -0.6458, 12Z 0.4544, 18Z 0.4365, 06-17 00Z -1.765, 06Z -2.342, 12Z -2.896
noaa_gfs	10-m northward wind (v_10m)	06-14 12Z 2.038, 18Z 4.411, 06-15 00Z 3.057, 06Z 2.524, 12Z 2.016, 18Z 3.091, 06-16 00Z -0.2301, 06Z 1.278, 12Z 1.881, 18Z 4.033, 06-17 00Z 0.5866, 06Z 0.9076, 12Z 0.4017
openaq	Ozone (ozone)	06-14 12Z 0.01561, 18Z 0.02208, 06-15 00Z 0.01393, 06Z 0.01061, 12Z 0.008989, 18Z 0.01508, 06-16 00Z 0.007862, 06Z 0.002926, 12Z 0.01025, 18Z 0.01699, 06-17 00Z 0.01466, 06Z 0.02004, 12Z 0.01982
openaq	PM10 (pm10)	06-14 12Z 33, 18Z 22.17, 06-15 00Z 33.06, 06Z 49.44, 12Z 41.22, 18Z 12.22, 06-16 00Z 20.89, 06Z 23.69, 12Z 18.33, 18Z 15, 06-17 00Z 18.89, 06Z 14.44, 12Z 13.67
openaq	PM2.5 (pm25)	06-14 12Z 7.408, 18Z 8.402, 06-15 00Z 9.636, 06Z 9.642, 12Z 7.469, 18Z 4.56, 06-16 00Z 7.97, 06Z 7.125, 12Z 4.252, 18Z 2.61, 06-17 00Z 2.806, 06Z 4.772, 12Z 4.544
openweather	Cloud cover (cloud_cover)	06-14 12Z 45.19, 18Z 81.5, 06-15 00Z 95.83, 06Z 99.86, 12Z 99.13, 18Z 99.93, 06-16 00Z 99.57, 06Z 99.33, 12Z 99.9, 18Z 100, 06-17 00Z 96.81, 06Z 75.69
openweather	Relative humidity (humidity)	06-14 12Z 84.92, 18Z 73.03, 06-15 00Z 89, 06Z 92.72, 12Z 93.48, 18Z 94.13, 06-16 00Z 92.93, 06Z 95.23, 12Z 94.17, 18Z 91.79, 06-17 00Z 91.1, 06Z 93.86
openweather	Precipitation rate (precipitation)	06-14 12Z 0.1485, 18Z 0.1917, 06-15 00Z 0.203, 06Z 0.19, 12Z 3.307, 18Z 4.336, 06-16 00Z 0.7973, 06Z 1.585, 12Z 1.259, 18Z 0.7517, 06-17 00Z 0.4219, 06Z 0.7976
openweather	Surface pressure (pressure)	06-14 12Z 1015, 18Z 1013, 06-15 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1012, 06-16 00Z 1012, 06Z 1011, 12Z 1011, 18Z 1009, 06-17 00Z 1008, 06Z 1006
openweather	Air temperature (temperature)	06-14 12Z 28.83, 18Z 31.85, 06-15 00Z 27.99, 06Z 27.09, 12Z 26.29, 18Z 25.43, 06-16 00Z 25.95, 06Z 25.36, 12Z 25.52, 18Z 26.41, 06-17 00Z 26.26, 06Z 25.04
openweather	Wind direction (wind_direction)	06-14 12Z 170, 18Z 182, 06-15 00Z 151.6, 06Z 164.7, 12Z 187.4, 18Z 163.7, 06-16 00Z 125.5, 06Z 198.5, 12Z 184.4, 18Z 165, 06-17 00Z 114.1, 06Z 111.2
openweather	Wind speed (wind_speed)	06-14 12Z 2.405, 18Z 3.247, 06-15 00Z 2.106, 06Z 2.097, 12Z 1.986, 18Z 1.815, 06-16 00Z 1.037, 06Z 1.571, 12Z 1.688, 18Z 2.229, 06-17 00Z 2.293, 06Z 2.014

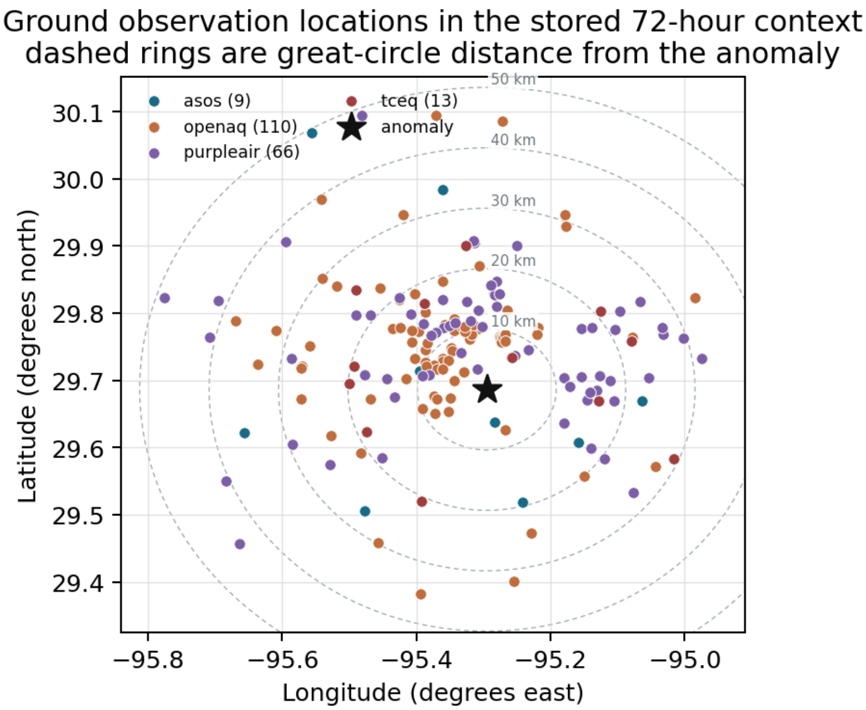
Source	Metric	Values
purpleair	PM2.5 (pm25)	06-14 12Z 8.86, 18Z 10.99, 06-15 00Z 11.83, 06Z 10.83, 12Z 7.916, 18Z 6.546, 06-16 00Z 9.512, 06Z 9.382, 12Z 4.645, 18Z 5.254, 06-17 00Z 4.878, 06Z 7.361, 12Z 7.826
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	CO granules available (s5p_co_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
tceq	Carbon monoxide (co)	06-14 12Z 0.2111, 18Z 0.1933, 06-15 00Z 0.2167, 06Z 0.1778, 12Z 0.2333, 18Z 0.2667, 06-16 00Z 0.2778, 06Z 0.2278, 12Z 0.2556, 18Z 0.2444, 06-17 00Z 0.2167, 06Z 0.1556, 12Z 0.2
tceq	Nitrogen dioxide (no2)	06-14 12Z 2.338, 18Z 1.518, 06-15 00Z 2.643, 06Z 2.819, 12Z 5.514, 18Z 7.59, 06-16 00Z 11.11, 06Z 8.731, 12Z 5.738, 18Z 7.23, 06-17 00Z 6.751, 06Z 4.889, 12Z 4.05
tceq	Sulfur dioxide (so2)	06-14 12Z -0.2333, 18Z -0.1278, 06-15 00Z -0.1833, 06Z -0.2389, 12Z -0.2111, 18Z -0.07222, 06-16 00Z 0.01667, 06Z -0.09444, 12Z -0.08333, 18Z -0.07222, 06-17 00Z 0.005556, 06Z -0.1, 12Z -0.1667

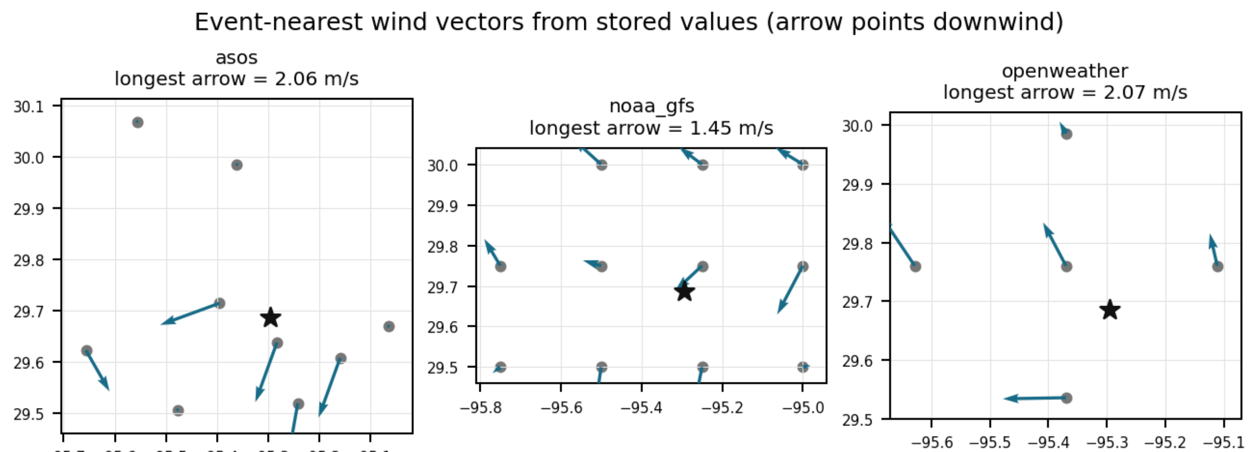
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	HOU (5.555 km) 87.63, MCJ (10.163 km) 92.6, EFD (15.811 km) 87.54, LVJ (19.307 km) 87.54, T41 (22.362 km) 87.58, AXH (26.678 km) 86.21, IAH (33.753 km) 91.33, SGR (35.675 km) 92.16, +1 more
asos	Air temperature (temperature)	HOU (5.555 km) 27.12, MCJ (10.163 km) 26.33, EFD (15.811 km) 27.25, LVJ (19.307 km) 26.96, T41 (22.362 km) 27.2, AXH (26.678 km) 26.59, IAH (33.753 km) 26.48, SGR (35.675 km) 26.54, +1 more
asos	Wind direction (wind_direction)	HOU (5.555 km) 136.8, MCJ (10.163 km) 125.7, EFD (15.811 km) 105.5, LVJ (19.307 km) 122.6, T41 (22.362 km) 147.9, AXH (26.678 km) 62.36, IAH (33.753 km) 135, SGR (35.675 km) 126.8, +1 more
asos	Wind speed (wind_speed)	HOU (5.555 km) 3.008, MCJ (10.163 km) 3.701, EFD (15.811 km) 3.318, LVJ (19.307 km) 2.315, T41 (22.362 km) 3.43, AXH (26.678 km) 1.079, IAH (33.753 km) 2.351, SGR (35.675 km) 2.701, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (8.297 km) 5888, gfs:29.75,-95.50 (21.05 km) 5889, gfs:29.50,-95.25 (21.162 km) 5889, gfs:29.50,-95.50 (28.689 km) 5890, gfs:29.75,-95.00 (29.33 km) 5888, gfs:30.00,-95.25 (35.148 km) 5888, gfs:29.50,-95.00 (35.231 km) 5889, gfs:30.00,-95.50 (40.109 km) 5888, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (8.297 km) 352.1, gfs:29.75,-95.50 (21.05 km) 373.5, gfs:29.50,-95.25 (21.162 km) 335.1, gfs:29.50,-95.50 (28.689 km) 303.2, gfs:29.75,-95.00 (29.33 km) 353.5, gfs:30.00,-95.25 (35.148 km) 357.2, gfs:29.50,-95.00 (35.231 km) 437.7, gfs:30.00,-95.50 (40.109 km) 398.2, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (8.297 km) 60.38, gfs:29.75,-95.50 (21.05 km) 59.7, gfs:29.50,-95.25 (21.162 km) 60.39, gfs:29.50,-95.50 (28.689 km) 60.32, gfs:29.75,-95.00 (29.33 km) 60.51, gfs:30.00,-95.25 (35.148 km) 59.67, gfs:29.50,-95.00 (35.231 km) 60.17, gfs:30.00,-95.50 (40.109 km) 59.07, +3 more

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (8.297 km) 1010, gfs:29.75,-95.50 (21.05 km) 1008, gfs:29.50,-95.25 (21.162 km) 1010, gfs:29.50,-95.50 (28.689 km) 1009, gfs:29.75,-95.00 (29.33 km) 1011, gfs:30.00,-95.25 (35.148 km) 1008, gfs:29.50,-95.00 (35.231 km) 1011, gfs:30.00,-95.50 (40.109 km) 1007, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (8.297 km) 18.3, gfs:29.75,-95.50 (21.05 km) 18.27, gfs:29.50,-95.25 (21.162 km) 18.38, gfs:29.50,-95.50 (28.689 km) 18.29, gfs:29.75,-95.00 (29.33 km) 18.32, gfs:30.00,-95.25 (35.148 km) 18.27, gfs:29.50,-95.00 (35.231 km) 18.49, gfs:30.00,-95.50 (40.109 km) 18.27, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (8.297 km) -0.7707, gfs:29.75,-95.50 (21.05 km) -1.022, gfs:29.50,-95.25 (21.162 km) -1.205, gfs:29.50,-95.50 (28.689 km) -1.28, gfs:29.75,-95.00 (29.33 km) -0.7837, gfs:30.00,-95.25 (35.148 km) -0.7484, gfs:29.50,-95.00 (35.231 km) -0.9445, gfs:30.00,-95.50 (40.109 km) -0.773, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (8.297 km) 1.655, gfs:29.75,-95.50 (21.05 km) 1.644, gfs:29.50,-95.25 (21.162 km) 2.206, gfs:29.50,-95.50 (28.689 km) 1.846, gfs:29.75,-95.00 (29.33 km) 2.177, gfs:30.00,-95.25 (35.148 km) 1.8, gfs:29.50,-95.00 (35.231 km) 3.429, gfs:30.00,-95.50 (40.109 km) 1.781, +3 more
openaq	Ozone (ozone)	325 (0.01 km) 0.01311, 3378 (6.399 km) 0.009662, 320 (7.255 km) 0.01229, 2039 (11.575 km) 0.01153, 1319333 (13.71 km) 0.01526, 2046 (15.8 km) 0.009925, 332 (16.16 km) 0.01793, 312 (18.679 km) 0.0142, +9 more
openaq	PM10 (pm10)	13380082 (6.399 km) 27.21, 15852419 (9.465 km) 27.4, 271 (16.16 km) 20.43
openaq	PM2.5 (pm25)	14140751 (0.011 km) 6.825, 15990898 (4.432 km) 5.912, 16564662 (4.852 km) 6.304, 13418566 (5.486 km) 6.499, 8967123 (6.375 km) 9.868, 8967479 (6.375 km) 8.683, 15539682 (6.38 km) 8.647, 3379 (6.399 km) 8.614, +82 more
openweather	Cloud cover (cloud_cover)	grid:center (10.975 km) 91.99, grid:south (18.24 km) 92.2, grid:east (19.553 km) 92.3, grid:west (33.267 km) 93.52, grid:north (33.994 km) 88.06
openweather	Relative humidity (humidity)	grid:center (10.975 km) 91.8, grid:south (18.24 km) 88.17, grid:east (19.553 km) 90.61, grid:west (33.267 km) 91.1, grid:north (33.994 km) 91.25
openweather	Precipitation rate (precipitation)	grid:center (10.975 km) 1.009, grid:south (18.24 km) 1.33, grid:east (19.553 km) 1.405, grid:west (33.267 km) 1.124, grid:north (33.994 km) 1.062
openweather	Surface pressure (pressure)	grid:center (10.975 km) 1011, grid:south (18.24 km) 1011, grid:east (19.553 km) 1012, grid:west (33.267 km) 1011, grid:north (33.994 km) 1012
openweather	Air temperature (temperature)	grid:center (10.975 km) 26.9, grid:south (18.24 km) 26.91, grid:east (19.553 km) 27.05, grid:west (33.267 km) 26.57, grid:north (33.994 km) 26.65
openweather	Wind direction (wind_direction)	grid:center (10.975 km) 158, grid:south (18.24 km) 155.3, grid:east (19.553 km) 166.2, grid:west (33.267 km) 158.8, grid:north (33.994 km) 160.6
openweather	Wind speed (wind_speed)	grid:center (10.975 km) 1.534, grid:south (18.24 km) 2.546, grid:east (19.553 km) 2.595, grid:west (33.267 km) 1.496, grid:north (33.994 km) 2.011
purpleair	PM2.5 (pm25)	6752 (3.679 km) 8.015, 99797 (6.962 km) 7.184, 133994 (7.18 km) 0.8767, 293725 (8.647 km) 10.35, 166435 (8.846 km) 10.44, 293735 (9.495 km) 10.54, 222601 (10.473 km) 5.223, 235425 (11.32 km) 8.752, +58 more
tceq	Carbon monoxide (co)	482011035 (6.378 km) 0.1352, 482011039 (16.162 km) 0.1333, 482011052 (16.841 km) 0.4043
tceq	Nitrogen dioxide (no2)	482010416 (0.0 km) 5.662, 482011035 (6.378 km) 8.662, 482011039 (16.162 km) 5.049, 482011052 (16.841 km) 14.91, 482011066 (19.515 km) 7.518, 482010055 (19.781 km) 5.327, 480391004 (20.723 km) 2.266, 482010026 (20.843 km) 7.172, +4 more
tceq	Sulfur dioxide (so2)	482011035 (6.378 km) -0.4192, 482011039 (16.162 km) 0.02466, 482010051 (18.688 km) 0.04384


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
asos (9)
openaq (110)
purpleair (66)
tceq (13)
anomaly
10 km
20 km
30 km
40 km
50 km
Latitude (degrees north)
Longitude (degrees east)


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 2.06 m/s
noaa_gfs
longest arrow = 1.45 m/s
openweather
longest arrow = 2.07 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

Concentrated primary nitrogen oxide emissions from evening traffic and industrial sources accumulated locally within a shallow boundary layer.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

TCEQ ground monitors recorded a severe NO_2 concentration anomaly of 18.2 ppb at 2026-06-16T00:00:00+00:00 in Houston relative to an expected baseline of 3.07 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

Coinciding with the NO2 peak at 2026-06-16T00:00:00Z, surface ozone measured across 17 OpenAQ sensors dropped to a 6-hour average of 0.00786 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

Ground-based monitoring confirmed a broad regional rise in 6-hour mean NO2 concentrations up to 11.11 ppb, centered on the anomaly monitor's 18.2 ppb peak.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

A sharp drop in planetary boundary layer height accompanied by weak surface wind speeds inhibited vertical and horizontal dispersion, trapping pollutants near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

On June 16, 2026 at 00:00 UTC, a severe NO_2 concentration anomaly of 18.2 ppb was recorded by a TCEQ monitor at coordinates (29.686, -95.295), significantly exceeding the expected baseline of 3.07 ppb with a z-score of 7.21.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

A nearby fresh combustion plume from Houston urban-roadway, industrial, marine, rail, or other fuel-burning activity is a likely direct cause of the severe NO_2 enhancement because the anomalous monitor measured 18.2 ppb while area TCEQ NO_2 was elevated at the same time and contemporaneous ozone at the collocated OpenAQ location was low at about 0.01 ppm, which is consistent with fresh NO_x influence near emission sources.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The location of the anomaly is in Houston, Texas, with a grid center point at (10.975 km) 91.8 and a distance of approximately 3.679 km from sensor 6752.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

The TCEQ Houston monitor at 29.686, -95.295 recorded a severe NO2 anomaly of 18.2 ppb at 2026-06-16 00:00 UTC against an expected value near 3.07 ppb, and the surrounding TCEQ network also showed elevated 6-hour mean NO2 at 00Z relative to earlier periods.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

Weak dispersion is a likely contributing cause of the severe NO2 enhancement because winds near the event were light at roughly 1 to 2 m/s and the modeled planetary boundary layer height was very shallow near the anomaly time, reaching about 48 m at the nearest GFS grid cell, which would favor trapping of primary NOx near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

The poor-dispersion accumulation hypothesis is strongly supported because meteorology near 2026-06-16T00:00:00Z featured very high humidity around 94 percent, light surface winds near 1 to 2 m/s, nearly complete cloud cover near 100 percent, and a very shallow modeled planetary boundary layer near 48 m, all of which favor trapping of NO2 close to sources.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

The pollutant anomaly is a TCEQ NO2 measurement of 18.2 ppb at 2026-06-16T00:00:00+00:00 at 29.686, -95.295 in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The chemical pattern near 2026-06-16 00:00 UTC is most consistent with a fresh local NOx emission plume because ozone was low while PM2.5, PM10, CO, and SO2 did not show comparably strong coincident spikes and satellite context does not indicate a broad regional pollution enhancement at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 8.7 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5 and NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

The combination of high PM2.5 and NO2 levels is a significant contributor to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

Moist, cloudy, nighttime conditions are a likely contributing cause of the severe NO2 enhancement because relative humidity was around 90 to 95 percent, cloud cover was near 100 percent, and ozone was depressed during the event, a combination that favors reduced photolysis and nocturnal chemical partitioning that can maintain emitted NOx as NO2 close to sources.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

The anomalous TCEQ NO2 value of 18.2 ppb is far above the expected value of 3.0676829268292685 ppb and has a z-score of 7.210558990276541, which is labeled severe in the anomaly record.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

ASOS weather stations and OpenWeather observations recorded stagnant surface wind speeds below 1.6 m/s and cloud cover near 100% at the time of the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

During the 6-hour period around 2026-06-16T00:00:00Z, the regional average NO2 concentration across 12 TCEQ monitors reached a period peak of 11.11 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

Planetary boundary layer heights dropping below 150 meters combined with light surface wind speeds near 1.5 m/s around 2026-06-16T00:00:00Z severely suppressed atmospheric dispersion.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

Cloud cover near 100% attenuated solar radiation, weakening photolysis rates and suppressing surface ozone formation while maintaining high ground-level NO2 concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

The anomaly occurred on June 16th, 2026, at approximately 00:00:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

NOAA GFS model output showed a collapse of the planetary boundary layer height down to 147.4 meters around 2026-06-16T00:00:00+00:00, severely restricting vertical atmospheric mixing.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

At 2026-06-16T00:00:00+00:00, nearby observations show openaq ozone of 0.01 ppm at the anomaly location and a 6-hour mean TCEQ NO2 of 11.11 ppb for 06-16 00Z, which is higher than the earlier 6-hour means of 7.59 ppb at 06-15 18Z and 5.514 ppb at 06-15 12Z.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

The broad regional or elevated-column enhancement hypothesis is weakened because particulate matter remained moderate near the anomaly, sulfur dioxide showed no concurrent increase, and the available Sentinel-5P trace-gas columns are either temporally distant from the event or do not show corroborating evidence for a large regional pollution episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

Houston-area meteorology at 2026-06-16 00:00 UTC was highly favorable for near-surface NOx accumulation because wind speeds were light, boundary-layer height was very low, humidity was high, and cloud cover was nearly complete.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

Overcast cloud cover reduced solar radiation and photolysis rates of nitrogen dioxide, enhancing its accumulation near the ground.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

The local fresh-combustion plume hypothesis is supported because the anomalous TCEQ monitor measured NO2 at 18.2 ppb versus an expected 3.07 ppb at 2026-06-16T00:00:00Z, while nearby ozone was only 0.01 ppm and nearby CO was modestly elevated to 0.3 ppm, a combination consistent with recently emitted NOx near the surface rather than a well-mixed aged regional plume.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

Industrial activities and vehicle traffic are likely contributing factors to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.